3D Printer



Category: **Computers, IT & Office** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: 3D Printer

What it is used for

A 3D printer creates three-dimensional objects by building material layer by layer from a digital model. Use the correct material, printer settings, and operating procedure to produce accurate parts while preventing burns, moving-part injuries, fumes, and electrical hazards.

Main parts

- Printer frame and build platform
- Print head, nozzle, or resin exposure unit, depending on technology
- Filament spool or resin/material feed system
- Motion rails, belts, and motors
- Control panel and display
- Power supply, enclosure, and ventilation system, if fitted

Before you start

1. Confirm the printer technology, material type, and intended print job.
2. Inspect the power cord, frame, platform, motion system, and protective covers.
3. Check that the nozzle, build surface, or resin vat is clean and correctly installed.
4. Use material approved for the printer and review its safety data where applicable.
5. Prepare the digital model and verify orientation, supports, dimensions, and slicer settings.
6. Ensure the printer is on a stable surface with adequate ventilation and clear space around moving parts.

How to use

1. Switch off the printer before installing material or performing manual adjustments that require access to internal parts.
2. Load filament or prepare the resin/material system according to the manufacturer's instructions.
3. Secure the build platform and check its level or calibration as required.
4. Transfer the prepared print file using the approved method and verify the selected material and settings.
5. Close the enclosure or protective cover where required and start the print job.
6. Observe the first layers or initial build stage to confirm proper adhesion and material flow.
7. Monitor progress through the display or approved remote monitoring system.
8. If a print failure, unusual noise, smoke, or material leak occurs, stop the printer when safe and follow the fault procedure.

9. Allow the build platform, nozzle, resin, or printed part to cool or be handled according to the material instructions.
10. Remove the finished part using the approved tool and avoid touching hot or sharp components.
11. For resin printers, use the specified washing and post-curing process with appropriate PPE.
12. Clean the build area, remove waste material safely, and record any maintenance or print issues.

⚠ Safety

- Only trained users should operate the printer and perform maintenance.
- Hot nozzles, heated beds, and cured or uncured materials can cause burns or irritation; follow material-specific precautions.
- Keep fingers, hair, loose clothing, and tools away from moving axes and belts.
- Provide appropriate ventilation, especially for resin, heated plastics, and materials that release fumes.
- Use gloves and eye protection when handling resin, solvents, sharp parts, or uncured prints.
- Disconnect power before electrical servicing; never bypass interlocks or protective covers.

Care & storage

- Clean the build platform and nozzle using the manufacturer's approved procedure after cooling.
- Remove dust and material residue from rails, fans, and accessible surfaces.
- Inspect belts, fasteners, cables, fans, and motion components regularly.
- Replace worn consumables and filters according to the maintenance schedule.
- Store filament or resin according to its manufacturer's moisture, temperature, and light requirements.

Common mistakes to avoid

- Starting a print without checking platform calibration or material settings.
- Touching a hot nozzle or moving mechanism during operation.
- Printing in a poorly ventilated area without material-specific precautions.
- Removing a print with excessive force or handling uncured resin without protection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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